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Concept Review

Thermocouples

Why Use Thermocouples?

Temperature measurement plays an essential role in many sectors, ranging from automotive and manufacturing to healthcare to consumer products. Thermocouples are widely used in industrial temperature sensing, selected for their versatility, ruggedness, and cost-effectiveness. Real-world applications range from monitoring jet engine exhaust to controlling plastic injection molding with precision accuracy. Understanding their operating principles, types, and specifications helps us to select the right sensor and ensure proper application.

Thermocouples

A thermocouple is a simple temperature sensing device that measures the voltage generated across two different materials in response to temperature changes. They consist of two wire conductors that are joined at one end to form an electrical junction. Thermocouples can be constructed in varying forms and using different materials, as shown in Figure 1 below:

Figure 1: Different forms of thermocouples. Image source: https://www.realpars.com/blog/thermocouple

A thermocouple leverages the **Seebeck effect**, which occurs when a temperature gradient causes the flow of electrons within a material. The main components of a thermocouple are shown in Figure 2. The **hot junction**, where the wires are joined, is exposed to the medium being measured. The wires remain separated at the other end, called the **cold junction** or **reference point**, typically connected to a measuring device. The temperature difference between the two junctions causes electrons to flow. Since the two wires are made of different materials, the electrons flow at different rates, creating a small potential difference (voltage) signal. The voltage, proportional to the temperature difference between the junctions, is measured at the cold junction and then converted to a temperature measurement.

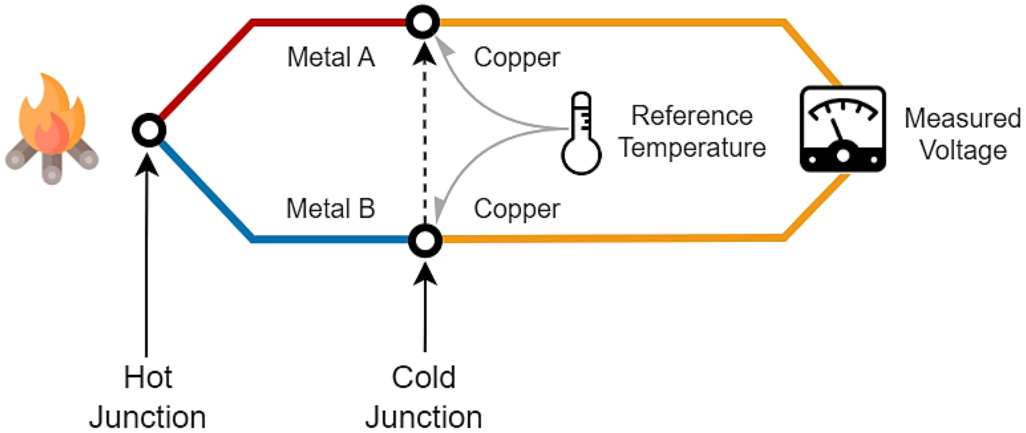


Figure 2: Diagram of a thermocouple. Source: https://erqos.com/thermocouples-vs-rtds-choosing-the-right-temperature-sensor-for-your-application/

Cold Junction Compensation

A key takeaway is that a thermocouple by itself only measures the **differential temperature** between the hot junction and the cold junction. To determine **absolute temperature**, a separate sensor is required to measure the temperature of the cold junction. The cold junction ideally remains at a constant temperature and is often placed close to the measuring device. The reference temperature of the cold junction is used to calculate the absolute temperature through a process called **cold junction compensation (CJC)**.

In modern thermocouple systems, a semiconductor temperature sensor (such as the MAX31855 interface chip located in the Mechatronic Sensors Trainer) measures the ambient temperature at the cold junction. The system then measures the voltage from the thermocouple output at the reference junction, converts this to an uncompensated temperature value and adds the cold junction temperature to calculate the actual hot junction temperature.

This compensation happens in dedicated thermocouple interface chips, making implementation with microcontrollers quick and straightforward. The MAX31855, for example, features a 14-bit ADC, offering 0.25˚C resolution with ±2˚C accuracy for K-type thermocouples, reading temperatures between -200˚C and +700˚C.

Types of Thermocouples

Different pairs of metals produce different voltage-temperature characteristics. Table 1 summarizes common thermocouple types, designated by standardized letters that define their specific combination of metals and their corresponding properties. Type K is frequently used due to its wide temperature range, good accuracy, reliability, and relatively low cost.

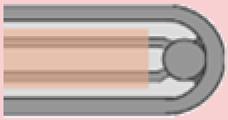
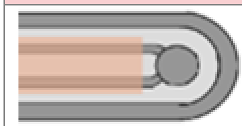
Table 1: Summary of common thermocouple types

Type	Metal Pairing	Temperature Range	Characteristics
K	Chromel / Alumel	-200˚C to +1,260˚C	General purpose, most popular
J	Iron / Constantan	-40˚C to +750˚C	Plastics, vacuum applications (not for moist environments)
T	Copper / Constantan	-200˚C to +350˚C	Low temperature, cryogenic processes
N	Nicrosil / Nisil	-270˚C to +1,300˚C	High-temperature stability, oxidation resistance
E	Chromel / Constantan	-270˚C to +1,000˚C	High output, good accuracy
R, S	Platinum-Rhodium alloys	0˚C to +1,600˚C	High-precision, expensive

Junction Styles

When the thermocouple wires and their junction are inserted into a protective metal tube called a sheath (commonly stainless steel), it is called a **thermocouple probe**. The wires inside the sheath do not touch except at the measuring junction. There are 3 types of ways in which the junction and the sheath can come together, summarized in Table 2:

Table 2: Summary of junction styles

Junction Type	Description	Considerations
 Grounded	The sheath and wires are welded together at the junction	Fast response time but susceptible to noise from electrical interference
 Ungrounded	The wire junction is insulated from the sheath	Highest accuracy but slowest response time
 Exposed	The junction is unprotected and directly contacts the medium	Fastest response but susceptible to corrosion and oxidation

You can verify if your thermocouple is grounded or ungrounded by using a voltmeter to check for conductivity between the sheath and one of the 2 leads on the connector end. If there is conductivity, it has a grounded junction.

Datasheet And Considerations

When selecting a thermocouple, key requirements are determined by several factors such as the operating environment, measuring medium, and measurement range. Detailed specifications for a specific thermocouple model can be found in the datasheet. A few considerations are summarized below:

Temperature range: the measurement range is influenced by multiple factors, of which a major one is the thermocouple type. Some types, such as T, are selected for low temperatures, while others, like type R or S, are more suitable for high temperature applications.

Response time: this indicates how fast the measurement changes to reflect a change in temperature. Response time is affected by the probe diameter (larger diameter = slower response), the wire gauge (thicker wire = slower response) and the junction style (exposed < grounded < ungrounded). This specification is often expressed as a time constant, defined as the time required to reach 63.2% of a step change in temperature under specified conditions.

Environmental conditions: the operating environment may require the thermocouple to have resistance to chemical, vibration, abrasion, and pressure. Ensure that the material properties of various thermocouple elements such as the sheath, junction style, and wire insulation can meet these requirements.

Comparison with Other Temperature Sensors

Other sensors that measure temperature include thermistors, which are temperature-sensitive resistors, and resistance temperature detectors (RTDs), which are sensors that use the known resistance-temperature relationship of a specific material. The thermocouple has many advantages in being a rugged, inexpensive, self-powered sensor with a wide measurement range. However, their main limitation is accuracy, where achieving system errors of less than one degree Celsius can be challenging. Thermistors are another inexpensive option that work well for precise measurements but only within a small range. RTDs offer the highest accuracy and stability as the most expensive option.

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